

SULAYMANIYAH, Iraq

WMO: 406230

Lat: 35.550N

Lon: 45.450E

Elev: 853

StdP: 91.49

Time Zone: 3.00 (E03)

Period: 89-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.2	-2.9	-12.0	1.5	7.0	-9.9	1.8	8.2	10.4	8.3	9.4	8.9	1.7	300	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	43.0	19.3	41.9	19.2	40.9	18.7	21.2	39.1	20.3	38.2	19.6	37.6	4.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	12.5	22.3	14.9	11.7	21.3	13.9	11.0	21.3	65.3	38.7	62.2	37.9	59.6	37.8	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.7	8.3	7.5		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	5.9	7.1	11.1	15.0	21.1	27.9	31.5	31.4	26.4	19.9	12.4	7.6
	DBStd	9.64	2.77	3.38	2.84	3.07	3.53	2.86	2.34	2.44	3.13	3.27	3.17	2.86
	HDD10.0	342	131	91	20	2	0	0	0	0	0	0	13	85
	HDD18.3	1580	387	315	225	108	13	0	0	0	1	21	177	334
	CDD10.0	3323	2	9	54	151	343	537	668	663	492	308	86	10
	CDD18.3	1520	0	0	1	7	97	287	409	405	243	71	1	0
	CDH23.3	24884	0	0	28	163	1593	4731	6967	6702	3632	1040	24	3
	CDH26.7	16290	0	0	3	26	749	3073	4963	4756	2293	426	1	0
(14) Wind	WSAvg	3.0	2.6	2.6	3.0	2.8	2.9	3.6	3.7	3.6	3.2	3.0	2.8	2.7
(15) Precipitation	PrecAvg	779	126	136	123	111	46	4	2	1	2	31	94	112
	PrecMax	1119	214	326	275	231	122	15	6	6	11	125	293	296
	PrecMin	445	37	67	24	13	2	0	0	0	0	4	11	11
	PrecStd	189	50	62	63	62	38	5	2	2	3	34	88	82
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.6	20.6	27.0	29.2	36.2	41.9	44.9	44.1	41.0	34.9	25.8	22.1
		MCWB	9.3	10.9	13.0	16.9	17.3	19.4	19.9	19.7	18.5	17.1	12.7	9.7
	2%	DB	14.9	18.2	23.0	27.0	34.8	40.2	43.2	43.0	39.8	32.9	23.9	17.9
		MCWB	8.5	10.2	12.8	15.4	17.2	18.4	19.3	19.3	17.6	16.5	11.5	8.5
	5%	DB	13.7	16.1	21.0	25.1	33.0	39.0	42.1	42.0	38.1	31.1	22.1	15.6
		MCWB	8.2	9.2	11.5	15.0	16.6	18.1	19.1	19.3	17.6	15.6	11.5	8.7
	10%	DB	12.1	14.2	19.0	23.1	31.2	37.8	40.9	41.0	36.3	29.2	20.0	13.8
		MCWB	7.5	8.5	11.2	14.6	16.3	17.7	18.5	19.1	17.1	15.0	11.0	8.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	12.8	15.6	18.7	20.3	21.0	22.5	22.4	20.6	19.1	16.0	12.5
		MCDB	13.8	17.1	21.7	26.3	29.1	38.5	41.4	41.0	37.2	31.6	21.6	14.7
	2%	WB	10.0	11.4	14.0	17.3	19.1	19.8	21.0	21.3	19.2	17.7	14.3	11.2
		MCDB	13.3	15.1	21.1	23.8	29.7	37.6	39.8	39.8	35.6	29.1	18.6	14.2
	5%	WB	9.1	10.2	12.9	16.2	18.2	19.0	20.1	20.4	18.4	16.7	13.3	10.1
		MCDB	11.9	14.3	18.4	22.7	29.0	37.0	39.0	39.1	35.2	26.9	17.9	13.1
	10%	WB	8.3	9.3	12.0	15.3	17.4	18.2	19.4	19.6	17.7	15.9	12.3	9.1
		MCDB	11.0	13.3	17.2	20.7	28.1	35.8	38.5	38.7	34.7	25.9	17.1	12.4
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	12.5	13.5	14.4	17.1	18.8	19.0	19.4	19.4	16.6	14.1	11.2
		MCDBR	14.7	16.3	18.1	18.2	20.1	20.6	20.6	20.2	20.6	19.7	18.0	15.3
		MCWBR	9.2	9.6	9.3	8.8	7.7	8.2	8.1	8.4	8.6	9.5	9.4	9.0
	5% WB	MCDBR	12.8	14.0	15.0	15.3	17.8	19.4	18.3	19.2	19.3	16.8	13.4	11.6
		MCWBR	8.6	8.9	8.5	8.1	7.8	8.3	8.0	8.4	8.7	8.7	7.7	7.7
(40) Clear-Sky Solar Irradiance	taub		0.370	0.408	0.458	0.540	0.533	0.470	0.499	0.461	0.419	0.443	0.382	0.365
	taud		2.147	2.046	1.933	1.773	1.808	1.956	1.891	1.985	2.079	2.005	2.162	2.175
	Ebn at Noon		801	812	804	759	771	820	793	816	832	763	780	778
	Edh at Noon		117	145	177	219	215	185	196	175	151	149	114	107
(44) All-Sky Solar Radiation	RadAvg		2.46	3.25	4.52	5.53	6.81	8.12	7.74	7.07	5.92	4.20	3.03	2.33
	RadStd		0.22	0.33	0.43	0.48	0.43	0.37	0.19	0.19	0.28	0.32	0.34	0.30

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page